Project Development Phase
Model Performance Test

Date	15 February 2026
Team ID	LTVIP2026TMIDS54303
Project Name	Online Payment Fraud Detection using Machine Learning
Maximum Marks	

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values
1	Model Summary	Multiple models trained: Random Forest, Decision Tree, Extra Trees, SVC Dataset Size: 16,426 balanced samples Features Used: step, amount, balances, transaction type (one-hot encoded)
2	Accuracy	**Random Forest:** Training Accuracy – 100% Test Accuracy – 99.27% **Decision Tree:** Training Accuracy – 100% Test Accuracy – 99.48% **Extra Trees:** Training Accuracy – 100% Test Accuracy – 98.81% **SVC:** Training Accuracy – 83.58% Test Accuracy – 83.26%
3	Confusion Matrix (Best Performing Model)	True Fraud Correctly Detected – 1610 False Fraud Alerts – 6–18 Precision ≈ 0.99 Recall ≈ 0.99 F1-Score ≈ 0.99
4	Fine Tuning Result	Decision Tree achieved highest Test Accuracy – **99.48%** Random Forest also performed strongly with **99.27%** accuracy